

# Sowdarjya Kolay

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## EDUCATION

### Academy of Technology

*B.Tech in Electronics & Communication Engineering*

2023 – 2027

*Kolkata, West Bengal*

### Heritage Academy High School

*Secondary & Higher Secondary*

2010 – 2023

*Kolkata, West Bengal*

## TECHNICAL SKILLS

**Languages:** Rust, TypeScript, JavaScript, Python, Java, Bash

**Systems/Backend:** Tokio (async runtime), Node.js, Express.js, FastAPI, REST APIs, Microservices

**Frontend:** Next.js, React.js, React Native, Redux, Tailwind CSS

**AI/ML:** LangChain, LangGraph, RAG, Vertex AI, OpenRouter, Pinecone

**Database & ORM:** PostgreSQL, MongoDB, Redis, Prisma ORM

**Cloud/DevOps:** Docker, AWS, GCP, Vercel, GitHub Actions, CI/CD

## EXPERIENCE

### BlockseBlock

Jul 2025 – Aug 2025

*Blockchain Developer Intern*

*Remote*

- Built a decentralized event-ticketing canister in Rust on the Internet Computer (DFX), exposing 13 endpoints across 4 core data types for a full on-chain ticket lifecycle.
- Implemented purchase logic enforcing sale windows, per-user ticket caps, and oversell prevention, with e8s-precision on-chain pricing and collision-resistant verification codes.
- Built a companion Vite/React frontend integrated with the canister via the DFINITY agent.

## PROJECTS

### Sparky — Redis-Compatible In-Memory Store | *Rust, Tokio* | GitHub

- Built a Redis-compatible in-memory data store in Rust: RESP2 TCP server with strings, lists, hashes, and sets, an actor-based single-owner design for deterministic command ordering, concurrent client handling, and AOF persistence with replay.
- Benchmarked head-to-head against Redis using **redis-benchmark** across 9 command types; Sparky sustains ~85% of Redis's throughput on average (~1.18× slower), with sub-millisecond p99 latency across all commands.
- Profiled and eliminated hot-path bottlenecks: switched database keys from **String** to **Bytes** to cut per-command allocation, added a reusable AOF write buffer, and batched actor command processing — improving average throughput from ~69% to ~85% of Redis.

### Wisp — Terminal AI Coding Agent | *TypeScript, Bun, OpenTUI* | GitHub

- Built a terminal-based AI coding agent with approval-gated tool-calling for file edits, command execution, and code search, backed by diff previews and persistent sessions.
- Verified correctness and safety with 152 automated tests covering tool execution, approval flow, session handling, and context limits.
- Enforced an approval-required-by-default policy for all destructive operations, with hard safeguards per command (2,000-char limit, 15s timeout, 1MB output cap) and a full audit log of every approval decision.

### D2P — AI-Powered GitHub PR Reviewer | *Next.js, TypeScript, Vertex AI, Prisma* | GitHub | Live

- Built an AI-powered GitHub PR review platform that scans diffs for secrets, CVEs, and OWASP patterns, then posts AI-verified findings directly to PRs with one-click fix-PR generation.
- Detects 8+ OWASP Top-10 vulnerability classes plus entropy-based secret detection (threshold 4.0), with CRITICAL findings automatically blocking merges.
- Scales CVE scanning via batched OSV queries (up to 1,000 packages/request, 60s timeout, cached) for fast checks on large monorepos.

## ACHIEVEMENTS & COMPETITIONS

### Hackathons & Competitions

- **2nd Place** — Gnosis Week, SCCSE-AOT (CTF)
- **3rd Place** — Decisia, Jadavpur University (Build & Pitch Competition)
- **Top 30 Finalist** — Hack4Bengal 4.0 (1000+ participants)